Human Robot Coordination

What would make this robot useful to you?

A robot can perform remarkable actions. Making them useful means agreeing what people need, in a particular situation.
Many everyday tasks need coordination before, during and after execution.

Supporting human-robot coordination
despite ~~based on~~ **generative AI**

Generative AI supports flexible dialogue.
Explicit state and control preserve the agreed task.

Agree. Act. Adapt.

A flexible workflow for human-robot coordination

Changed circumstances? Clarify and agree again.

1 Opening
Discover possibilities

HUMAN
What are you?
What can you do?
How can you make yourself useful?
How about you do ...?

→

2 Alignment
Clarify the task

HUMAN
Are there questions left unanswered?
Are there decisions I need to make?

→

3 Planning
Agree a plan

ROBOT
Here is what I can do.
Is that what you want?

HUMAN
Yes, start now.

→

4 Execution
Act and coordinate

ACTION
Carry out the agreed task.

HUMAN
Wait - something changed.

ROBOT
I need your help.

→

5 Reflection
Improve the next run

ROBOT
Was that OK?
What should change next time?

Speech and shared observations support every stage.

Agree changes before the next run.

Seven design guidelines

01 **Know and communicate capabilities.** Help people devise feasible tasks using the robot's abilities, available resources and limits.

02 **Make the workflow navigable.** Explain the current stage, suggest useful next steps and support explicit changes of direction.

03 **Make the agreement explicit.** Clarify goals, roles, conditions, limits and success.

04 **Keep commitments in state.** Preserve the agreement beyond the conversation.

05 **Ground meaning together.** Use speech, gestures and shared observations to clarify the task.

06 **Detect problems and support recovery.** Surface failures and uncertainty, explain their implications and seek human help when needed.

07 **Reflect and revise.** Agree specific changes before the next execution.

CURRENT CHALLENGE

Fast replies, faithful coordination

Natural conversation needs quick replies. Coordination also needs dependable meaning and control.

Beyond speech
Connect speech, visual cues and planned actions, including behaviour initiated by the robot.

Keep speech aligned
Keep spoken responses faithful to the current agreement as workflow stages and instructions change.

Check communication
Confirm meaning and check actual speech; transcripts alone cannot guarantee either.

Next: Measure and reduce response delays while preserving task control and checking spoken delivery.

Coordination in practice

Our Coordination Agent puts these guidelines into practice in PROMETHEUS.

01 / Explore and clarify

01 VALERIAN
I'm Valerian. We can explore what I can do, agree on a useful task, and try and improve it together. What would you like to achieve?

02 HUMAN
Who are you?

03 VALERIAN
I'm Valerian, a digital agent from ZHAW's SIRA Lab, built with PROMETHEUS. I help people explore and configure responsible multimodal interactions—though I'm still searching for Laureline, with no progress to report.

04 HUMAN
What can you do?

05 VALERIAN
I can converse, use supplied cues like visible counts or hand signs, and plan brief speech or expressive gestures. We can agree a bounded task, review it, then start it—you remain in control.

06 HUMAN
what does my face tell you?

07 VALERIAN
The supplied estimate labels your visible expression as happy, but it cannot tell me how you truly feel.

08 HUMAN
can you see this sign?

09 VALERIAN
When you asked, a fresh supplied hand observation reported scissors (recognition confidence 92%). This is a supplied estimate, not direct access to camera images.

02 / Agree, act and return

10 HUMAN
how about you do this: when I show a scissor sign you tell me something about your personal life, once.

11 VALERIAN
Proposed task: When a scissor sign is freshly recognised, I will share one brief fictional personal-life detail once.. Limit: 1 sensor action. Stop can end earlier. The proposal is awaiting your acceptance and is not active. After you accept, show a fresh matching cue; preparation signs do not count as task actions. Shall we start it now?

12 HUMAN
start now

13 VALERIAN
The task is active and waiting for a fresh matching cue. No task action has been performed yet.

14 VALERIAN
I once navigated by starlight and got gloriously lost; Laureline still had to rescue me.

15 HUMAN
do you miss laureline?

16 VALERIAN
Yes—though it's a fictional longing, and I keep hoping our paths cross again.

17 HUMAN
let's get back to conversation

18 VALERIAN
We are now in Conversation. Task execution has ended; our earlier conversation and recorded outcomes are preserved.

Complete recorded Valerian interaction. Wording and order preserved; typeset for print. The personal story is fictional.